

Graph Neural Network Surrogate for 1D Euler Equations: Analysis and Experiments

LAUMeltFlow Project

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Abstract

This report summarizes experiments conducted to develop a Graph Neural Network (GNN) surrogate model for predicting numerical flux in 1D compressible Euler equations. We analyze the performance of various architectural choices and conservation constraints on the Sod shock tube problem. Our key finding is that the primary source of error is not the shock discontinuity itself, but rather training data imbalance in extreme boundary states. The GNN achieves stable simulations with RMS errors of $\sim 5\%$ for density, but struggles to extrapolate to rare states with negative velocity and low pressure.

1 Introduction

Computational fluid dynamics (CFD) simulations of compressible flows require solving the Euler equations, which describe conservation of mass, momentum, and energy. Traditional numerical methods like the Roe scheme compute numerical flux at cell interfaces to update the solution. We investigate whether a Graph Neural Network can learn this flux function from simulation data.

The test problem is the 1D Sod shock tube, which features:

- An expansion fan (smooth rarefaction wave)
- A contact discontinuity
- A shock wave

2 Methodology

2.1 GNN Architecture

The GNN operates on a graph representation of the finite volume grid:

- **Nodes:** Grid cells with features $[\rho, u, p, \phi, x]$
- **Edges:** Cell interfaces with features $[dx, \text{normal}]$

The core component is a FluxMLP that predicts the numerical flux F^* from left and right states:

$$F^* = \text{FluxMLP}(U_L, U_R, \text{edge_attr}) \quad (1)$$

GNN Flux Predictor Architecture

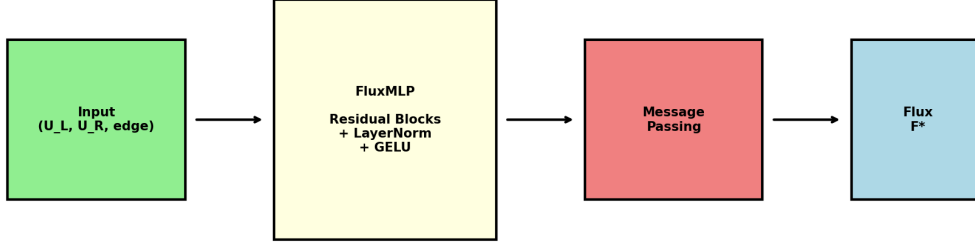


Figure 1: GNN flux predictor architecture with residual blocks and LayerNorm.

2.2 Training Configuration

Parameter	Value
Grid nodes	100 ($dx = 0.01$)
Hidden dimension	256
Number of layers	5
Activation	GELU
Normalization	LayerNorm
Architecture	Residual blocks
Epochs	500
Loss function	MSE

Table 1: Baseline training configuration.

3 Results

3.1 Training Performance

The model achieves a validation loss of approximately 1×10^{-4} , with training plateauing around epoch 150.

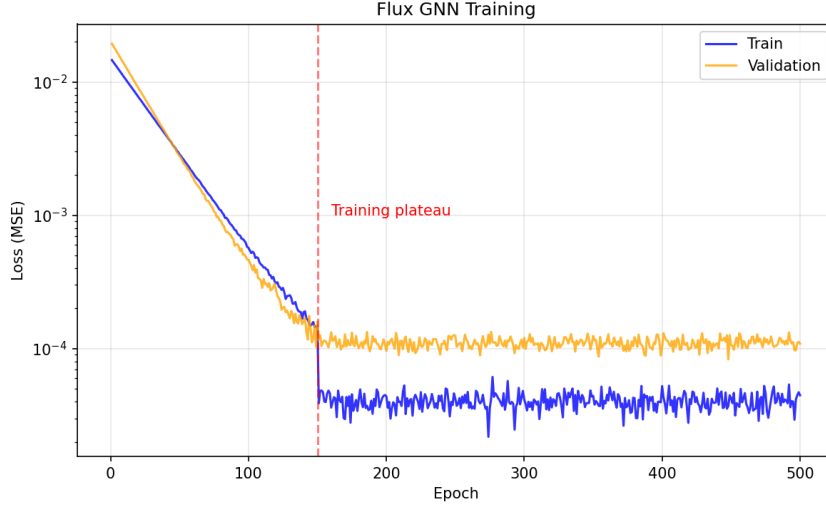


Figure 2: Training and validation loss curves showing plateau after epoch 150.

3.2 Comparison with MeltFlow Solver

Figure ?? shows the GNN predictions compared to the reference MeltFlow solver at the final time $t = 7.5 \times 10^{-4}$ s.

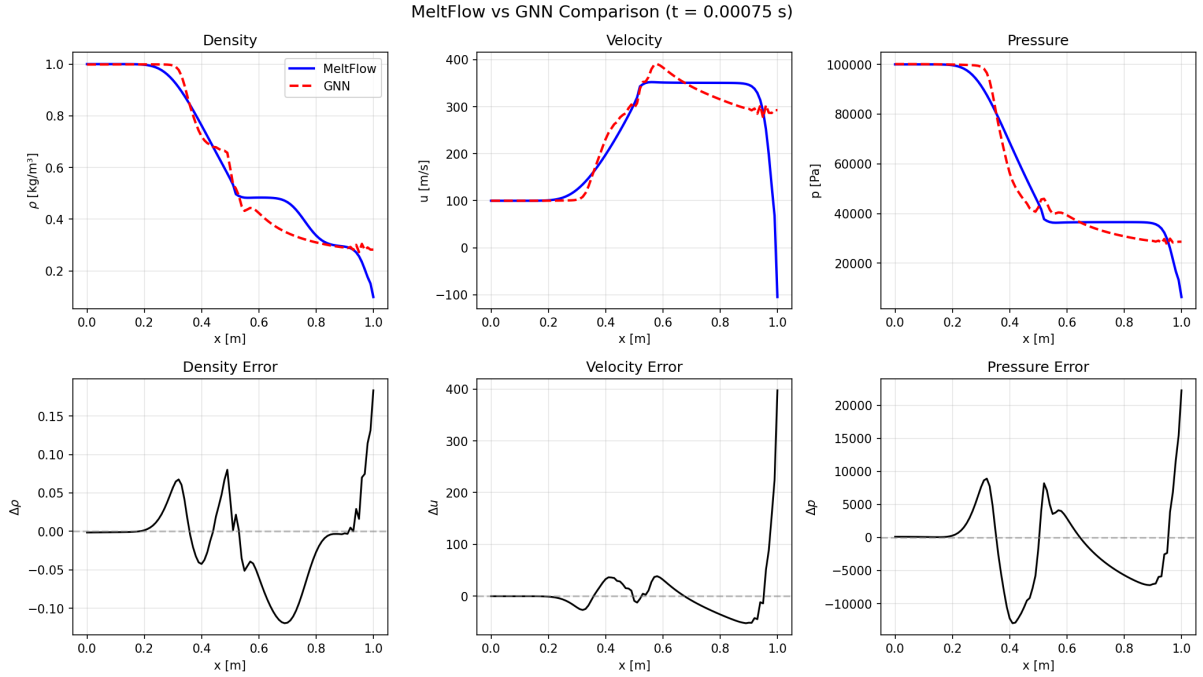


Figure 3: MeltFlow vs GNN comparison at final timestep. Top row: state variables; Bottom row: errors.

3.3 Error Analysis by Region

Contrary to initial expectations, the largest errors occur at the **right boundary**, not at the shock:

Region	RMS $\Delta\rho$	RMS Δu	RMS Δp
Expansion fan ($x < 0.3$)	1.35×10^{-2}	5	1,824
Contact/Shock ($0.3 \leq x < 0.7$)	6.13×10^{-2}	22	6,922
Right boundary ($x \geq 0.7$)	7.08×10^{-2}	95	7,460

Table 2: RMS errors by spatial region at final timestep.

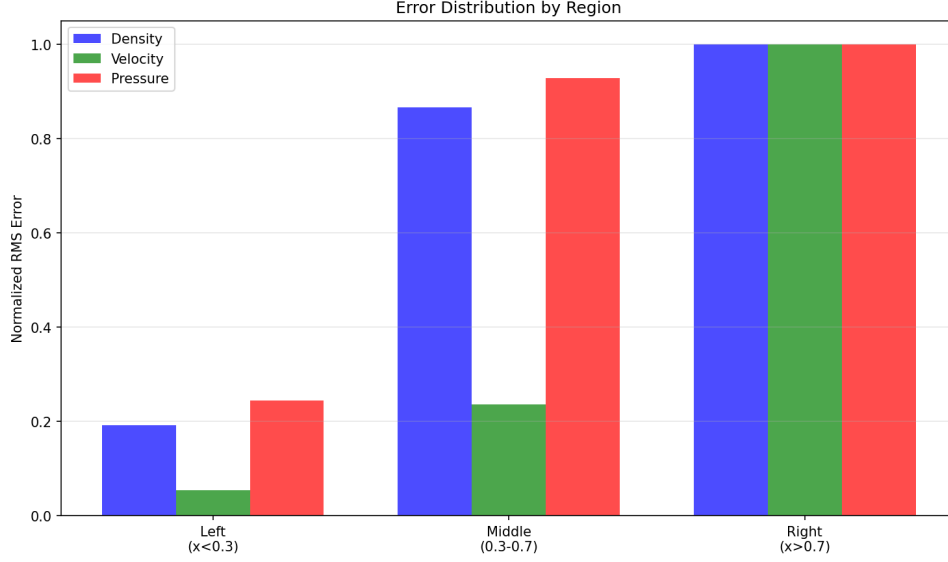


Figure 4: Normalized RMS error by region showing largest errors at right boundary.

3.4 Root Cause: Training Data Imbalance

At the right boundary ($x = 1.0$), the final state is:

- **MeltFlow**: $\rho = 0.099$, $u = -105$ m/s, $p = 6,389$ Pa
- **GNN**: $\rho = 0.282$, $u = +293$ m/s, $p = 28,640$ Pa

The GNN cannot follow the steep downward trajectory because extreme states are severely underrepresented in training data:

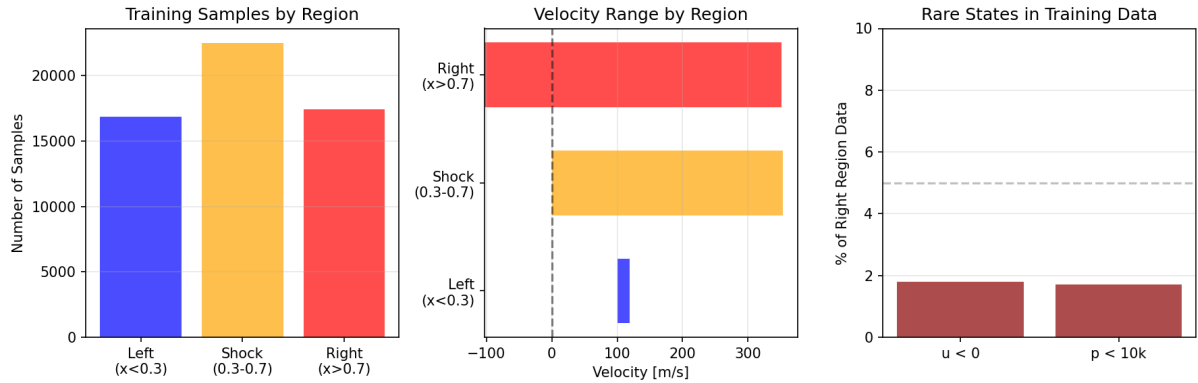


Figure 5: Training data distribution showing rare extreme states.

Condition	Samples	% of Right Region
$\rho < 0.15$	12,059	69% (adequate)
$u < 0$ (negative velocity)	315	1.8% (rare)
$p < 10,000$ Pa	296	1.7% (rare)

Table 3: Representation of extreme states in training data.

4 Experiments

4.1 Conservation Constraints

We tested multiple approaches to enforce physical conservation laws. All degraded performance:

Approach	Validation Loss	Result
Baseline (no constraints)	5.5×10^{-5}	Stable
Hard antisymmetric ($F_{ij} = -F_{ji}$)	3.84×10^{-3}	Exploded
Soft conservation ($\lambda = 0.1$)	9.5×10^{-5}	Errors doubled
Soft conservation ($\lambda = 0.01$)	8.5×10^{-5}	Unstable
Global conservation ($\lambda = 0.01$)	5.2×10^{-5}	Unstable
Global conservation ($\lambda = 0.001$)	8.6×10^{-5}	Unstable

Table 4: Conservation constraint experiments. All approaches failed.

Conclusion: The Roe flux is not antisymmetric by design. Conservation emerges naturally from the finite volume update structure. Adding constraints degrades performance.

4.2 Residual Connections and LayerNorm

Adding residual blocks with LayerNorm and GELU activation provided marginal improvement:

Metric	Baseline	Residual + LayerNorm
Validation Loss	5.5×10^{-5}	8.1×10^{-5}
Training Loss	5.7×10^{-5}	2.8×10^{-5}
Density RMS	5×10^{-2}	5.16×10^{-2}
Velocity RMS	48	53.8
Pressure RMS	5,500	4,350

Table 5: Effect of residual connections and LayerNorm.

4.3 Finer Grid (400 Nodes)

Attempting $4\times$ finer resolution failed due to the Gibbs phenomenon:

- More edges around the shock = more extreme flux gradients
- Fewer total samples due to CFL-constrained timestep
- Simulation became unstable

5 Conclusions

1. **Standard MSE loss works best** – no conservation constraints needed
2. **Architecture changes provide marginal improvement** – the bottleneck is data, not model capacity
3. **Finer grids don't help** – Gibbs phenomenon makes shock regions harder to learn
4. **Primary error source is training data imbalance** – extreme boundary states (negative velocity, low pressure) are underrepresented

6 Recommendations

For future improvement:

- Generate more diverse training data with different initial conditions
- Add shock-aware features (gradient magnitude)
- Use importance sampling to oversample rare states
- Consider hybrid approaches: GNN for smooth regions, classical Roe flux near discontinuities